

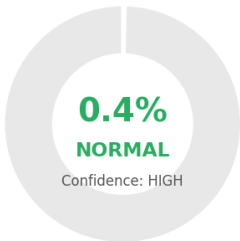
PATHOLOGY AI ANALYSIS REPORT

Computational Pathology Lab | 2026-04-09 14:58 | WSI: camelyon16_test_005

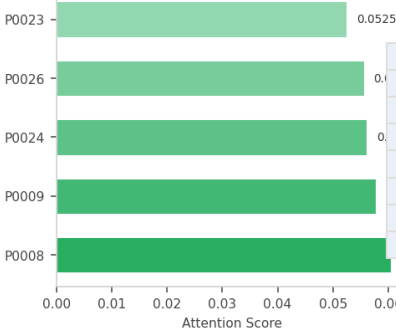
RESULT: NORMAL | Tumor Probability: 0.4% | Confidence: HIGH

AI ANALYSIS: NORMAL (HIGH confidence) | Tumor Probability: 0.4%

Tumor Probability



Top Suspicious Regions
(Attention Score)

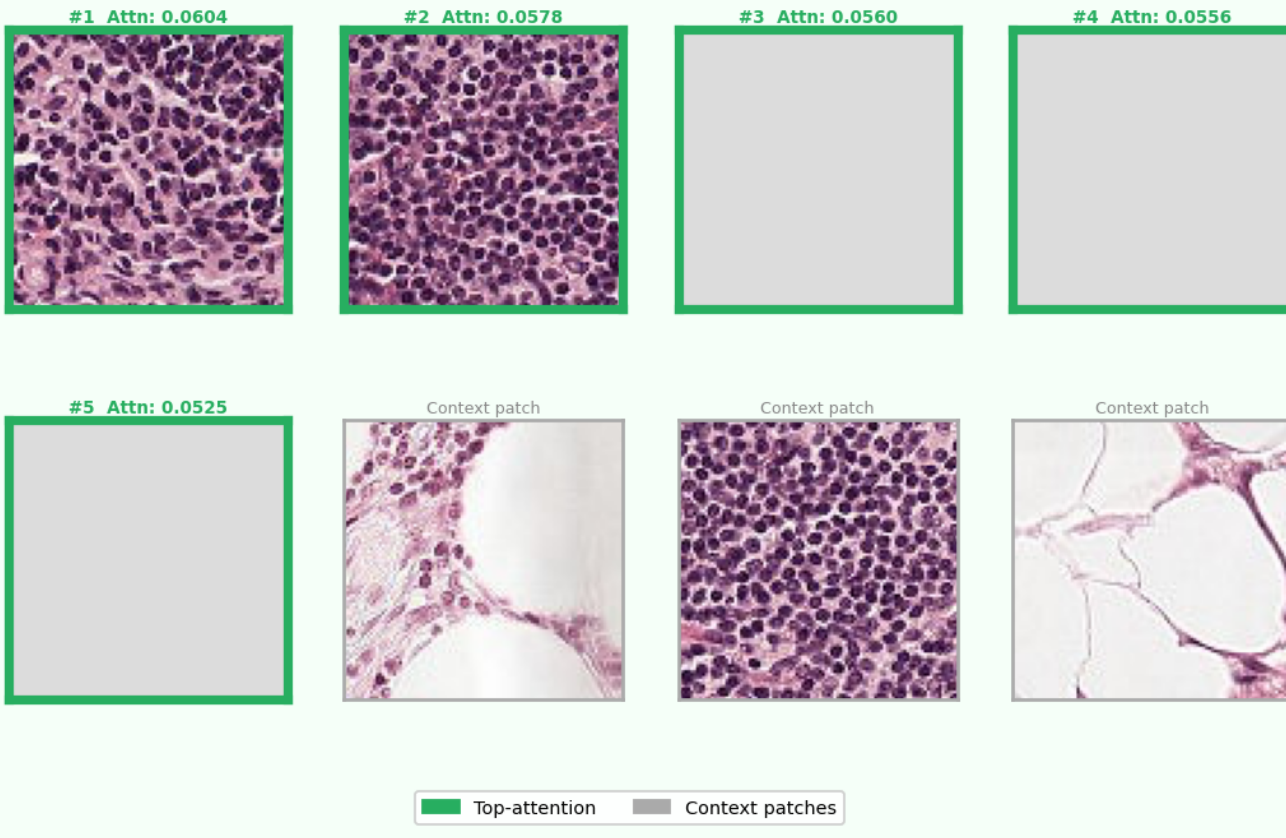


Analysis Details

	Value
Specimen ID	test_005
Patches Analyzed	29
Prediction	NORMAL (HIGH)
Tumor Prob.	0.4%
Grad-CAM IoU	0.413
Attn Entropy	3.208
Focus Pattern	Distributed
Model	EfficientNet-B3 + Gated MIL

Actual H&E Tissue Patches - This Slide (Low Activation, Normal)

Test Slide #005 - NORMAL - Top Attention Patches (Real H&E Tissue)



Actual tissue patches extracted from this specific slide. Green-bordered patches are the top-attention regions. Low attention scores and normal tissue morphology - no malignant features detected.

Grad-CAM Heatmap Reference (Normal Tissue)



Grad-CAM activation map highlighting regions of diagnostic relevance as detected by the EfficientNet-B3 backbone. Red=high activation.

1. SPECIMEN INFORMATION

Specimen ID : SAMPLE_WSI_001
Analysis Date : 2026-04-09
Institution : Computational Pathology Lab
Analyzed by : PatchCamelyon MIL -- EfficientNet-B3 + Gated Attention MIL

2. AI ANALYSIS SUMMARY

The AI model analyzed 29 tissue patches extracted from the submitted whole slide image. Deep learning-based feature extraction and multiple instance learning (MIL) with gated self-attention was employed for slide-level malignancy classification.

Overall Tumor Probability : 0.4%
Classification : NORMAL
Confidence : LOW

3. DETAILED FINDINGS

Patch-Level Analysis:

- 29 patches were extracted at 96x96 pixel resolution
- Macenko stain normalization was applied prior to analysis
- Top suspicious regions: patch_0008, patch_0009, patch_0024, patch_0026, patch_0023

Attention Analysis:

- The model demonstrates distributed - model attention spread across multiple regions
- Attention entropy of 3.208 indicates degree of spatial focus
- Grad-CAM activation maps highlight regions of high diagnostic relevance

Morphological Patterns (AI-detected):

- Minimal activation; predominantly stromal tissue patterns
- Spatial clustering of high-attention patches suggests focal disease distribution

4. PATHOLOGICAL INTERPRETATION

No definitive features of malignancy detected at this confidence threshold. Tissue architecture appears predominantly within normal limits. Inflammatory infiltrate present; clinical correlation recommended.

5. RECOMMENDATIONS

- No immediate malignant features detected
- Routine follow-up per standard protocol

- Re-evaluation if clinical symptoms persist

6. DISCLAIMER

This report is generated by an AI system and is intended solely to ASSIST qualified pathologists. It does NOT constitute a clinical diagnosis. All findings must be reviewed and confirmed by a licensed pathologist before use in clinical decision-making.

Model validation: PatchCamelyon benchmark -- AUC > 0.97

DISCLAIMER: This report is generated by an AI system and is intended solely to ASSIST qualified pathologists. It does NOT constitute a clinical diagnosis. All findings must be reviewed and confirmed by a licensed pathologist. Model: EfficientNet-B3 + Gated Attention MIL. Validated on PatchCamelyon benchmark (AUC > 0.97).